ELITE IGCSE ACADEMY

Private Past Paper Solutions

May 2023 P1HR Worked Solutions

May 2023 | P1HR

31 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May 2023 P1HR - May 2023 | P1HR

1 Write 2250 as a product of powers of its prime factors.
Show your working clearly.

....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$2250 = 225 \times 10$$

$$225 = 3^2 \times 5^2, \qquad 10 = 2 \times 5$$

So

$$2250 = 3^2 \times 5^2 \times 2 \times 5$$

$$2250 = 2 \times 3^2 \times 5^3$$

FINAL ANSWER

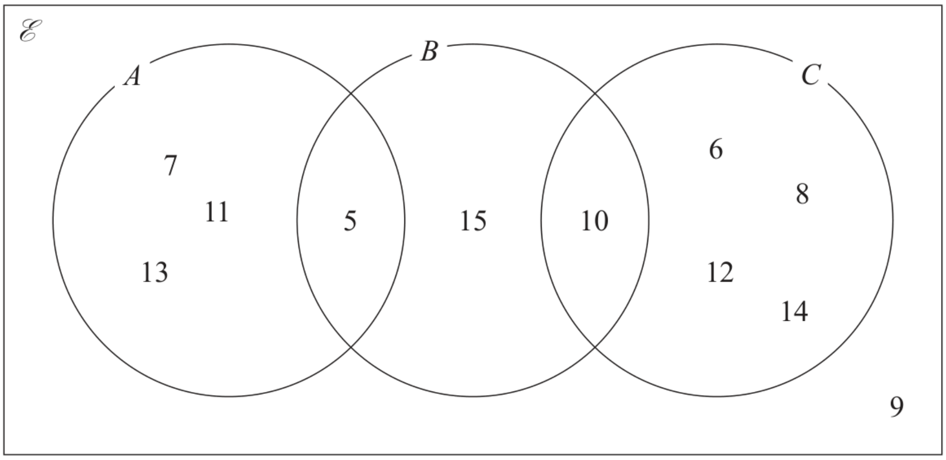
$2 \times 3^2 \times 5^3$

PAST PAPER QUESTION Q#: 2 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

May 2023 P1HR - May 2023 | P1HR

2 Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

....

(1)

(ii) $B \cup C$

....

(1)

Dominic writes down $9 \notin C$

(b) Explain why Dominic is correct.

....

....

(1)

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

The numbers inside A are the numbers in the A-only region and the overlap with B:

$$A = \{5, 7, 11, 13\}$$

$B \cup C$ means every number inside B or C:

$$B \cup C = \{5, 6, 8, 10, 12, 14, 15\}$$

Dominic writes $9 \notin C$. This is correct because 9 is outside circle C in the Venn diagram.

FINAL ANSWER

(a)(i) $\{5, 7, 11, 13\}$, (a)(ii) $\{5, 6, 8, 10, 12, 14, 15\}$. Dominic is correct because 9 is not in C.

PAST PAPER QUESTION

Q#: 3 Marks: 4

Topic: **Congruence, Similarity & Geometrical Proof**

May 2023 P1HR - May 2023 | P1HR

3 *ABCD* and *EFGH* are similar quadrilaterals.

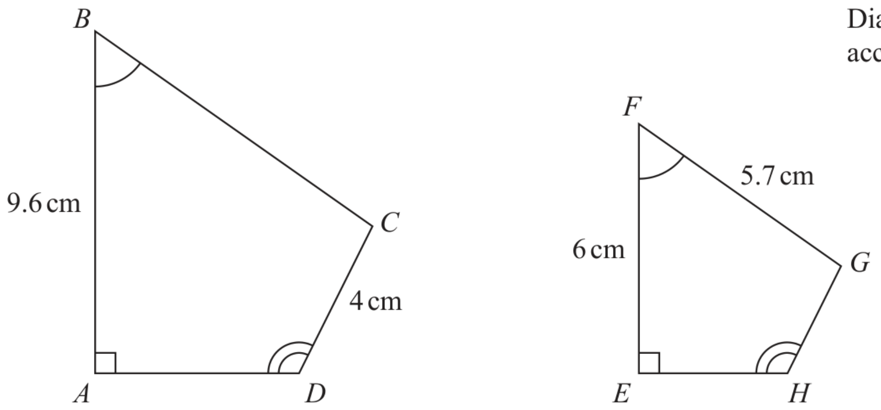

Diagram **NOT** accurately drawn

(a) Work out the length of *GH*

.... cm

(2)

(b) Work out the length of *BC*

.... cm

(2)

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

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Topic check. Congruence, Similarity & Geometrical Proof. The tag is correct.

Solution

The scale factor from $EFGH$ to $ABCD$ is

$$\frac{9.6}{6} = 1.6$$

Since GH corresponds to CD,

$$GH = \frac{4}{1.6} = 2.5$$

Since FG corresponds to BC,

$$BC = 5.7 \times 1.6 = 9.12$$

FINAL ANSWER

$GH = 2.5$ cm, and $BC = 9.12$ cm.

PAST PAPER QUESTION

Q#: 4 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1HR - May 2023 | P1HR

4 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

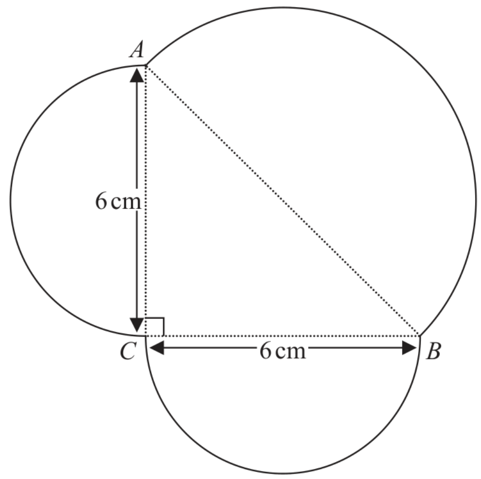

Diagram **NOT** accurately drawn

AB, BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

.... cm

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

TOPIC: **Circles, Arcs & Sectors**

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Q#: 4 Marks: 5

WORKED SOLUTION

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Topic check. Corrected to Circles, Arcs & Sectors.

Solution

$$AB = \sqrt{6^2 + 6^2} = \sqrt{72} = 8.485\ldots$$

The perimeter is the sum of three semicircular arcs:

$$\frac{\pi}{2}(AB + BC + CA) = \frac{\pi}{2}(8.485\ldots + 6 + 6) = 32.178\ldots$$

FINAL ANSWER

32.2 cm.

PAST PAPER QUESTION

Topic: **Probability Toolkit**

May 2023 P1HR - May 2023 | P1HR

Q#: 5 Marks: 2

5 Each time Evie plays a game against her computer, she will win or lose.

For each game, the probability that Evie will win is 0.74
Evie is going to play 300 games against her computer.

Work out an estimate for the number of games that Evie will lose.

....

(Total for Question 5 is 2 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 2

TOPIC: **Probability Toolkit**

May 2023 P1HR - May 2023 | P1HR

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Topic check. Probability Toolkit. The tag is correct.

Solution

$$P(\text{lose}) = 1 - 0.74 = 0.26$$

Expected losses:

$$300 \times 0.26 = 78$$

FINAL ANSWER

78.

PAST PAPER QUESTION Q#: 6 Marks: 4

TOPIC: **Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

6 (a) Simplify $m^{10} \div m^3$

....
(1)

$k^n \times k^4 = k^{12}$

(b) Write down the value of n

$n =$ ....
(1)

(c) Simplify $(3x^6y^8)^2$

....
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Algebraic Roots & Indices**

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Topic check. Algebraic roots and indices. The tag is correct.

Solution

(a) $m^{10} \div m^3 = m^{10-3} = m^7$.
(b) $k^n \times k^4 = k^{n+4} = k^{12}$, so $n + 4 = 12$ and $n = 8$.
(c) $(3x^6y^8)^2 = 3^2x^{12}y^{16} = 9x^{12}y^{16}$.

FINAL ANSWER
(a) m^7, (b) 8, (c) $9x^{12}y^{16}$

PAST PAPER QUESTION

Q#: 7 Marks: 3

TOPIC: **Factorising**

May 2023 P1HR - May 2023 | P1HR

7 (a) Expand $4x(x - 5)$

....

(1)

(b) Factorise $y^2 - 9y + 20$

....

(2)

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Factorising**

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WORKED SOLUTION

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Topic check. Corrected to factorising. The original tag was expanding brackets, but factorising carries more marks.

Solution

(a)

$$4x(x-5) = 4x^2 - 20x$$

(b)

$$y^2 - 9y + 20 = (y-4)(y-5)$$

FINAL ANSWER

(a) $4x^2 - 20x$, (b) $(y-4)(y-5)$

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

May 2023 P1HR - May 2023 | P1HR

8 (a) Write 5.6×10^{-3} as an ordinary number.

....

(1)

(b) Work out $\dfrac{6 \times 10^{3}}{2.1 \times 10^{-4} + 9 \times 10^{-5}}$

Give your answer in standard form.

....

(2)

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$5.6 \times 10^{-3} = 0.0056$$

For part (b), simplify the denominator:

$$2.1 \times 10^{-4} + 9 \times 10^{-5} = 0.00021 + 0.00009$$

$$= 0.00030 = 3 \times 10^{-4}$$

Then

$$\frac{6 \times 10^{3}}{3 \times 10^{-4}} = 2 \times 10^{3-(-4)}$$

$$= 2 \times 10^{7}$$

FINAL ANSWER

(a) 0.0056, (b) 2×10^{7}

PAST PAPER QUESTION

Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2023 P1HR - May 2023 | P1HR

9 Kazi buys a car for 700 000 taka.
The value of the car depreciates by 12% each year.

Work out the value of the car at the end of 3 years.
Give your answer correct to the nearest taka.

.... taka

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

A depreciation of 12% gives multiplier

$$0.88$$

After 3 years,

$$700000(0.88)^3 = 477030.4$$

Correct to the nearest taka,

$$477030.4 \approx 477030$$

FINAL ANSWER

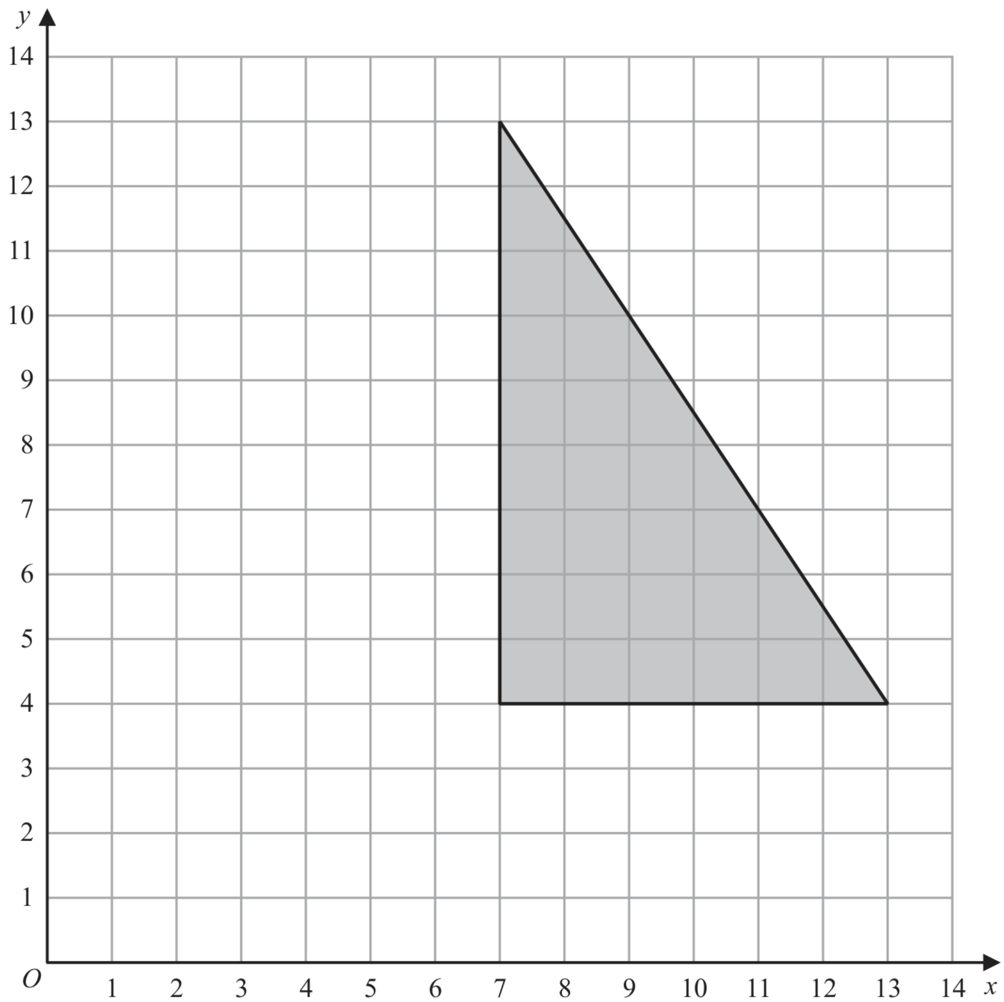
477030 taka

PAST PAPER QUESTION Q#: 10 Marks: 2

TOPIC: **Transformations**

May 2023 P1HR - May 2023 | P1HR

10



On the grid, enlarge the shaded shape with scale factor $\frac{1}{3}$ and centre (1, 7)

(Total for Question 10 is 2 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 2

TOPIC: **Transformations**

May 2023 P1HR - May 2023 | P1HR

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Topic check. Transformations. The tag is correct.

Solution

Use

$$\text{new point} = (1,7) + \frac{1}{3}(\text{old point} - (1,7))$$

The original vertices are

$$(7,4),\ (13,4),\ (7,13)$$

They become

$$(3,6),\ (5,6),\ (3,9)$$

FINAL ANSWER

draw the triangle with vertices $(3,6),(5,6),(3,9)$.

PAST PAPER QUESTION Q#: 11 Marks: 3

TOPIC: **Standard & Compound Units**

May 2023 P1HR - May 2023 | P1HR

11 The diagram shows a block of iron in the shape of a cuboid.

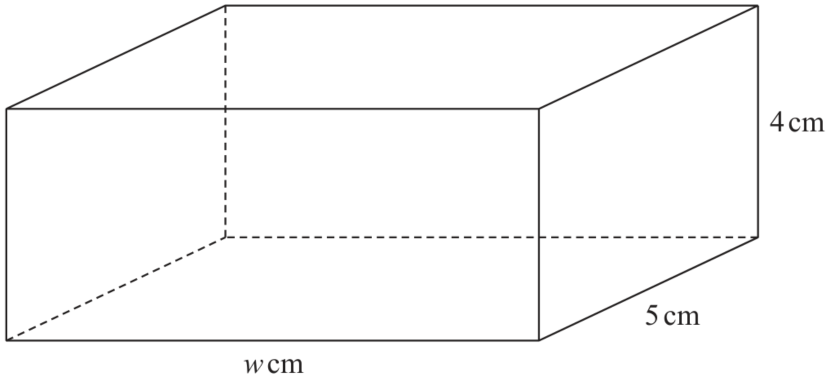

Diagram **NOT** accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g/cm^3
The mass of the block is 1950 g

Work out the value of w

w = ...

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Standard & Compound Units**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Corrected to Standard & Compound Units.

Solution

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$7.8 = \frac{1950}{\text{volume}}$$

$$\text{volume} = \frac{1950}{7.8} = 250 \text{ cm}^3$$

For the cuboid,

$$w \times 5 \times 4 = 250$$

$$20w = 250$$

$$w = 12.5$$

FINAL ANSWER

$w = 12.5$ cm.

PAST PAPER QUESTION

Q#: 12 Marks: 5

TOPIC: **Combined & Conditional Probability**

May 2023 P1HR - May 2023 | P1HR

12 Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	x	0.15	0.5	y	0.13	0.03

(a) Show that $x + y = 0.19$

(2)

Given that $3x - y = 0.09$

and $x + y = 0.19$

(b) work out the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 5

TOPIC: **Combined & Conditional Probability**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Combined & Conditional Probability. The tag is correct.

Solution

The probabilities add to 1:

$$x + 0.15 + 0.50 + y + 0.13 + 0.03 = 1$$

$$x + y = 0.19$$

Given

$$3x - y = 0.09$$

Add the two equations:

$$4x = 0.28$$

$$x = 0.07$$

$$y = 0.19 - 0.07 = 0.12$$

FINAL ANSWER

$x = 0.07,\ y = 0.12.$

PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

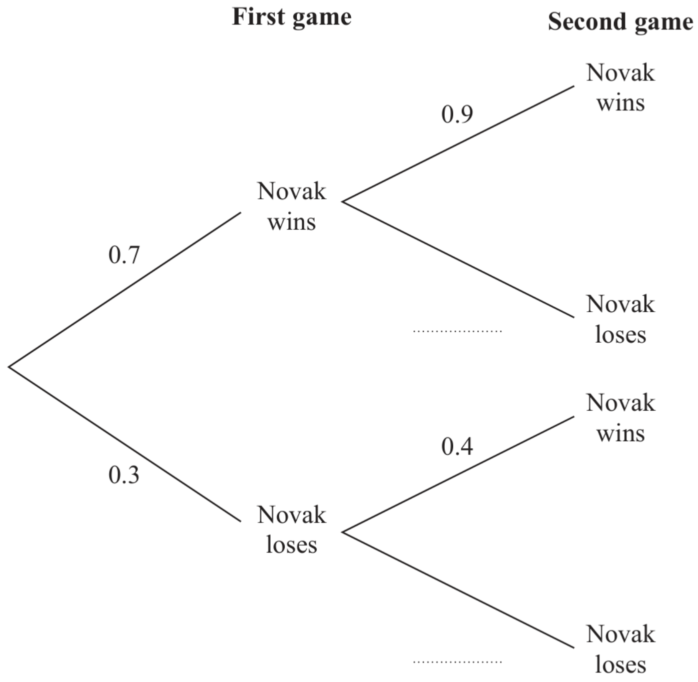
May 2023 P1HR - May 2023 | P1HR

13 Novak is going to play two games of tennis.

The probability that he wins the first game is 0.7

If he wins the first game, the probability that he wins the second game is 0.9
If he loses the first game, the probability that he wins the second game is 0.4

(a) Complete the probability tree diagram.



(1)

(b) Work out the probability that Novak wins both games of tennis.

....

(2)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

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WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

The missing branch probabilities are

$$0.1, \quad 0.6$$

Novak wins both games with probability

$$0.7 \times 0.9 = 0.63$$

FINAL ANSWER

0.63.

PAST PAPER QUESTION

Q#: 14 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

May 2023 P1HR - May 2023 | P1HR

14 The table gives information about the times taken by 80 people to run a race.

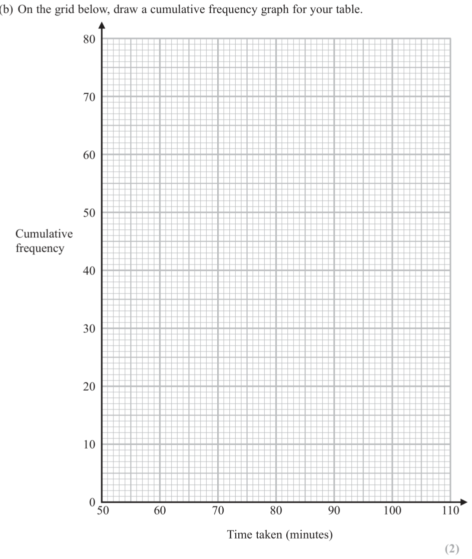
Time taken (t minutes)	Frequency
$50 < t \leqslant 60$	15
$60 < t \leqslant 70$	16
$70 < t \leqslant 80$	21
$80 < t \leqslant 90$	14
$90 < t \leqslant 100$	8
$100 < t \leqslant 110$	6

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$50 < t \leqslant 60$	
$50 < t \leqslant 70$	
$50 < t \leqslant 80$	
$50 < t \leqslant 90$	
$50 < t \leqslant 100$	
$50 < t \leqslant 110$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time taken.

.... minutes

(1)

(d) Use your graph to find an estimate for the interquartile range of the times taken.

.... minutes

(2)

(Total for Question 14 is 6 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

The cumulative frequencies are

$$15,\ 31,\ 52,\ 66,\ 74,\ 80$$

For 80 runners,

$$\text{median} = 40, \qquad Q_1 = 20, \qquad Q_3 = 60$$

$$\text{median} = 70 + \frac{40-31}{52-31} \times 10 \approx 74.3$$

$$Q_1 = 60 + \frac{20-15}{31-15} \times 10 \approx 63.1$$

$$Q_3 = 80 + \frac{60-52}{66-52} \times 10 \approx 85.7$$

$$\text{IQR} \approx 85.7 - 63.1 = 22.6$$

FINAL ANSWER

median about 74.3 min, IQR about 22.6 min.

PAST PAPER QUESTION

Q#: 15 **Marks: 4**

TOPIC: **Angles in Polygons & Parallel Lines**

May 2023 P1HR - May 2023 | P1HR

15

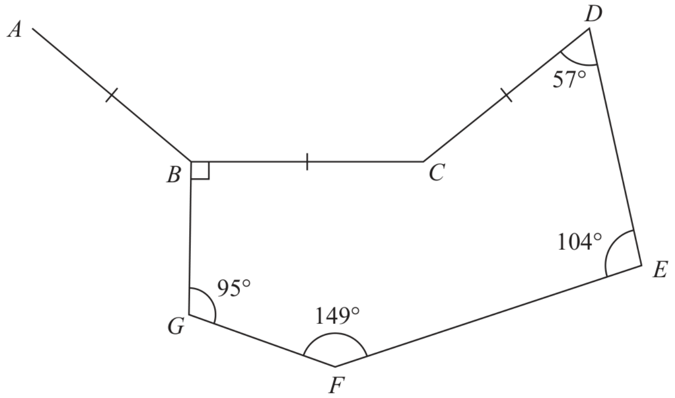


Diagram **NOT** accurately drawn

BCDEFG is a hexagon.
AB, *BC* and *CD* are three sides of a regular *n*-sided polygon.

Calculate the value of *n*
Show your working clearly.

$n = $ ....

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

The sum of the interior angles of a hexagon is

$$(6-2)\times 180 = 720^\circ$$

The reflex angle at C in the hexagon is

$$720^\circ - (90^\circ + 95^\circ + 149^\circ + 104^\circ + 57^\circ) = 225^\circ$$

So the smaller angle BCD is

$$360^\circ - 225^\circ = 135^\circ$$

This is the interior angle of the regular n-sided polygon.
So the exterior angle is

$$180^\circ - 135^\circ = 45^\circ$$

$$n = \frac{360}{45} = 8$$

FINAL ANSWER

$n = 8$.

PAST PAPER QUESTION

Q#: 16 **Marks: 2**

TOPIC: **Algebraic Proof**

May 2023 P1HR - May 2023 | P1HR

16 Use algebra to show that $0.1\dot{7}\dot{6} = \dfrac{35}{198}$

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION Q#: 16 Marks: 2

TOPIC: **Algebraic Proof**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is correct.

Solution

Let

$$x = 0.1767676\ldots$$

Then

$$10x = 1.767676\ldots$$

and

$$1000x = 176.767676\ldots$$

Subtract:

$$990x = 175$$

$$x = \frac{175}{990} = \frac{35}{198}$$

FINAL ANSWER

$0.1767676\ldots = \dfrac{35}{198}$.

PAST PAPER QUESTION

Q#: 17 Marks: 4

TOPIC: **Direct & Inverse Proportion**

May 2023 P1HR - May 2023 | P1HR

17 F is inversely proportional to the square of r

$F = 36$ when $r = 4$

(a) Find a formula for F in terms of r

....
(3)

(b) Work out the value of F when $r = 48$

....
(1)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Direct & Inverse Proportion**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Direct and Inverse Proportion. The tag is correct.

Method

F is inversely proportional to r^2, so

$$F = \frac{k}{r^2}$$

Use $F = 36$ when $r = 4$:

$$36 = \frac{k}{4^2}$$

$$k = 36 \times 16 = 576$$

So

$$F = \frac{576}{r^2}$$

When $r = 48$,

$$F = \frac{576}{48^2} = \frac{576}{2304} = 0.25$$

FINAL ANSWER

$F = \frac{576}{r^2}$, and $F = 0.25$ when $r = 48$.

PAST PAPER QUESTION Q#: 18 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May 2023 P1HR - May 2023 | P1HR

18 Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.
The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.
Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.
Show your working clearly.

£..

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The fence length 50 m correct to the nearest 5 m has bounds

$$47.5 \leq L < 52.5$$

The panel length 1.3 m correct to the nearest 10 cm has bounds

$$1.25 \leq p < 1.35$$

Smallest possible number of panels:

$$\left\lceil \frac{47.5}{1.35} \right\rceil = 36$$

Greatest possible number of panels:

$$\left\lceil \frac{52.5}{1.25} \right\rceil = 42$$

Difference in number of panels:

$$42 - 36 = 6$$

Greatest difference in cost:

$$6 \times 8.65 = 51.90$$

FINAL ANSWER

51.90 pounds

PAST PAPER QUESTION

Q#: 19 Marks: 5

TOPIC: **Vectors**

May 2023 P1HR - May 2023 | P1HR

19 The diagram shows a triangle ABC where A, B and C represent the positions of three towns.

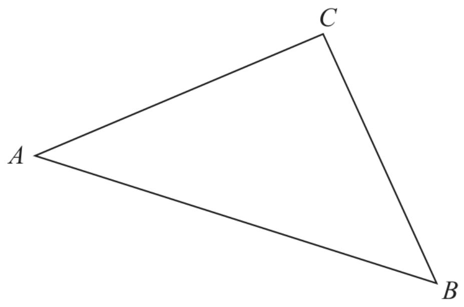

Diagram **NOT** accurately drawn

$$\overrightarrow{AB} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} \qquad \overrightarrow{BC} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

Pru travels directly from A to B and then directly from B to C

Yang travels directly from A to C

Given that the values for $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are in kilometres,

work out how much further Pru travels than Yang travels.
Give your answer in km, correct to one decimal place.

.... km

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Vectors**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Corrected to Vectors.

Solution

Pru travels

$$|AB| + |BC| = \sqrt{7^2 + (-2)^2} + \sqrt{(-3)^2 + 5^2}$$

$$= \sqrt{53} + \sqrt{34} = 13.111\ldots$$

Yang travels from A to C.

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$|AC| = \sqrt{4^2 + 3^2} = 5$$

Difference:

$$13.111\ldots - 5 = 8.111\ldots$$

FINAL ANSWER

Pru travels 8.1 km further.

PAST PAPER QUESTION

Q#: 20 Marks: 7

TOPIC: **Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

20 Curve **C** has equation $y = f(x)$
The graph of curve **C** has one maximum point.
The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.......................... , ...)
(1)

(ii) $y = f(x) - 7$

(.......................... , ...)
(1)

(iii) $y = f(-x)$

(.......................... , ...)
(1)

Curve **L** has equation $y = x^2 + 7x + 20$

Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

(b) Find an equation for **S**
Give your answer in the form $y = ax^2 + bx + c$

$y =$ ....
(4)

(Total for Question 20 is 7 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 7

TOPIC: **Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

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Topic check. Correct.

Method

The maximum point of $y = f(x)$ is $(3, 5)$.
For

$$y = 2f(x)$$

the y-coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

For

$$y = f(x) - 7$$

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

For

$$y = f(-x)$$

the graph reflects in the y-axis:

$$(3, 5) \rightarrow (-3, 5)$$

For part (b),

$$y = x^2 + 7x + 20$$

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

So the graph moves 2 units right.
Replace x by $x - 2$:

$$y = (x-2)^2 + 7(x-2) + 20$$

$$y = x^2 + 3x + 10$$

FINAL ANSWER

$(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

PAST PAPER QUESTION

Q#: 20 Marks: 7

TOPIC: **Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

20 Curve **C** has equation $y = f(x)$
The graph of curve **C** has one maximum point.
The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.......................... , ...)
(1)

(ii) $y = f(x) - 7$

(.......................... , ...)
(1)

(iii) $y = f(-x)$

(.......................... , ...)
(1)

Curve **L** has equation $y = x^2 + 7x + 20$

Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

(b) Find an equation for **S**
Give your answer in the form $y = ax^2 + bx + c$

$y =$ ....
(4)

(Total for Question 20 is 7 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 7

TOPIC: **Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The maximum point of $y = f(x)$ is $(3, 5)$.
For

$$y = 2f(x)$$

the y-coordinate is multiplied by 2:

$$(3, 5) \to (3, 10)$$

For

$$y = f(x) - 7$$

the graph moves 7 units down:

$$(3, 5) \to (3, -2)$$

For

$$y = f(-x)$$

the graph reflects in the y-axis:

$$(3, 5) \to (-3, 5)$$

For part (b),

$$y = x^2 + 7x + 20$$

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

So the graph moves 2 units right.
Replace x by $x - 2$:

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

FINAL ANSWER
$(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

PAST PAPER QUESTION

Q#: 21 Marks: 6

TOPIC: **Differentiation**

May 2023 P1HR - May 2023 | P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = $ ....

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

....

(4)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Differentiation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

Since the quadratic opens upwards, it is positive outside the roots.

FINAL ANSWER

(a) $\frac{dy}{dx} = 3x^2 - 4x - 9$, (b) $x < -1.19$ or $x > 2.52$

PAST PAPER QUESTION

Q#: 21 Marks: 6

TOPIC: **Differentiation**

May 2023 P1HR - May 2023 | P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} =$ ....

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

....

(4)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Differentiation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

Since the quadratic opens upwards, it is positive outside the roots.

FINAL ANSWER

(a) $\frac{dy}{dx} = 3x^2 - 4x - 9$, (b) $x < -1.19$ or $x > 2.52$

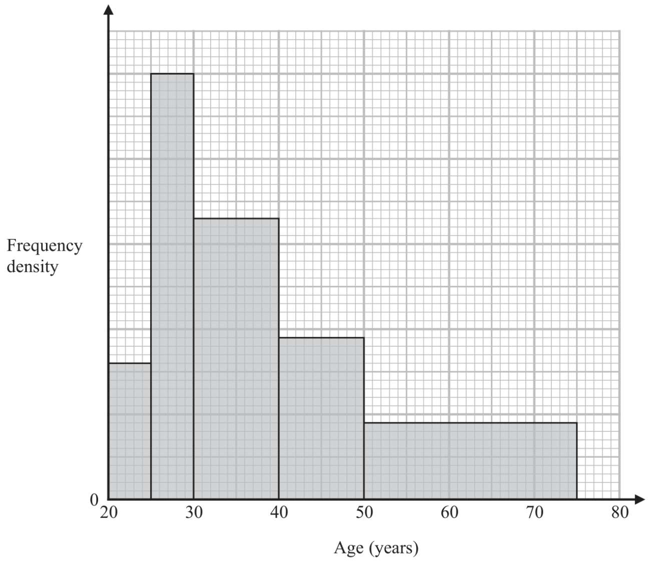
PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Histograms**

May 2023 P1HR - May 2023 | P1HR

22 Some people attend a concert.
The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

....%

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Histograms**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The scale of the frequency density axis is not needed because the percentage is found from areas. Using the bar heights from the histogram, the relative areas are:

$$(5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155)$$

$$= 1385 + 4330 + 5710 + 3290 + 3875$$

$$= 18590$$

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675\ldots$$

FINAL ANSWER

16.7%

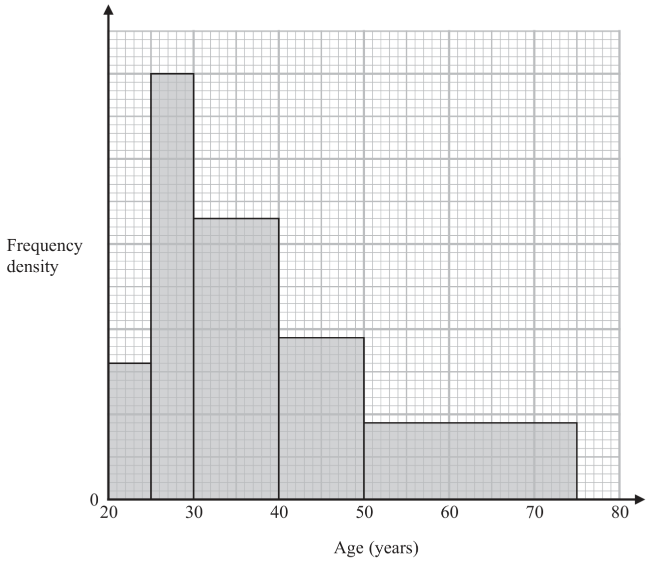
PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Histograms**

May 2023 P1HR - May 2023 | P1HR

22 Some people attend a concert.
The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

..............................%

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 **Marks: 4**

TOPIC: **Histograms**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The scale of the frequency density axis is not needed because the percentage is found from areas. Using the bar heights from the histogram, the relative areas are:

$$(5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155)$$

$$= 1385 + 4330 + 5710 + 3290 + 3875$$

$$= 18590$$

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$$(20)(155) = 3100$$

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675\ldots$$

FINAL ANSWER

16.7%

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

23 Simplify $(x^2 - 4) \div \left(\dfrac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\dfrac{ax^2}{bx + c}$ where a, b and c are integers.

....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Surds to Algebraic Fractions.

Method

$$(x^2-4) \div \left(\frac{4x^2-7x-2}{x}\right) - 2x$$

Change division into multiplication by the reciprocal:

$$(x^2-4) \times \frac{x}{4x^2-7x-2} - 2x$$

Factorise:

$$x^2-4=(x-2)(x+2)$$

$$4x^2-7x-2=(4x+1)(x-2)$$

So

$$\frac{x(x-2)(x+2)}{(4x+1)(x-2)} - 2x = \frac{x(x+2)}{4x+1} - 2x$$

$$\frac{x(x+2)-2x(4x+1)}{4x+1} = \frac{x^2+2x-8x^2-2x}{4x+1} = \frac{-7x^2}{4x+1}$$

FINAL ANSWER

$$\frac{-7x^2}{4x+1}$$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

23 Simplify $\left(x^2 - 4\right) \div \left(\dfrac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\dfrac{ax^2}{bx + c}$ where a, b and c are integers.

....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Surds to Algebraic Fractions.

Method

$$(x^2-4) \div \left(\frac{4x^2-7x-2}{x}\right) - 2x$$

Change division into multiplication by the reciprocal:

$$(x^2-4) \times \frac{x}{4x^2-7x-2} - 2x$$

Factorise:

$$x^2-4=(x-2)(x+2)$$

$$4x^2-7x-2=(4x+1)(x-2)$$

So

$$\frac{x(x-2)(x+2)}{(4x+1)(x-2)} - 2x = \frac{x(x+2)}{4x+1} - 2x$$

$$\frac{x(x+2)-2x(4x+1)}{4x+1} = \frac{x^2+2x-8x^2-2x}{4x+1} = \frac{-7x^2}{4x+1}$$

FINAL ANSWER

$$\frac{-7x^2}{4x+1}$$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic roots and indices. The tag is correct.

Solution

$$16^x = 2^{4x}, \qquad 8 = 2^3$$

So

$$2^n = 2^{x^2} \times 2^{4x} \times 2^3 = 2^{x^2+4x+3}$$

Hence

$$n = x^2 + 4x + 3 = (x+2)^2 - 1$$

$$(x+2)^2 = n + 1$$

Since $x > 0$,

$$x + 2 = \sqrt{n+1}$$

$$x = -2 + \sqrt{n+1}$$

For $x > 0$, $\sqrt{n+1} > 2$, so $n > 3$.

FINAL ANSWER

$x = -2 + \sqrt{n+1}$, $n > 3$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$2^n = 2^{x^2} \times 16^x \times 8$$

$$16^x = (2^4)^x = 2^{4x}$$

$$8 = 2^3$$

So

$$2^n = 2^{x^2+4x+3}$$

$$n = x^2 + 4x + 3$$

Complete the square:

$$n = (x+2)^2 - 1$$

$$(x+2)^2 = n + 1$$

Since $x > 0$, take the positive root:

$$x + 2 = \sqrt{n+1}$$

$$x = \sqrt{n+1} - 2$$

For $x > 0$,

$$\sqrt{n+1} > 2$$

$$n > 3$$

FINAL ANSWER

$x = \sqrt{n+1} - 2$, with $n > 3$

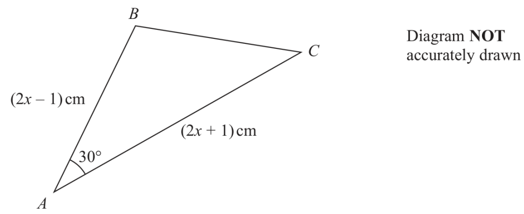
PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

25 Here is a triangle *ABC*



The area of the triangle is $(x^2 + x - 3.75)\,\text{cm}^2$

Find the size of the largest angle in triangle *ABC*
Give your answer correct to the nearest degree.

.... °

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Sine, Cosine Rule & Area of Triangles.

Method

The two sides around the 30° angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

Using the area formula,

$$\text{area} = \frac{1}{2}ab\sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x-1)(2x+1)\sin 30^\circ$$

Since $\sin 30^\circ = \frac{1}{2}$,

$$x^2 + x - 3.75 = \frac{1}{4}(2x-1)(2x+1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

So

$$AB = 2x - 1 = 6, \qquad AC = 2x + 1 = 8$$

Use the cosine rule to find BC:

$$BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 30^\circ$$

$$BC = 4.1063\ldots$$

The largest side is $AC = 8$, so the largest angle is $\angle B$.
Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\ldots^\circ$$

FINAL ANSWER

$\boxed{103^\circ}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

25 Here is a triangle *ABC*

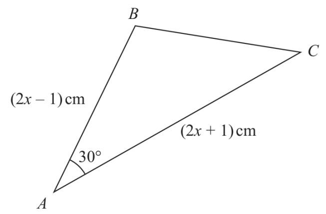


Diagram **NOT** accurately drawn

The area of the triangle is $(x^2 + x - 3.75)\,\text{cm}^2$

Find the size of the largest angle in triangle *ABC*
Give your answer correct to the nearest degree.

.... °

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Sine, Cosine Rule & Area of Triangles.

Method

The two sides around the 30° angle are

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Using the area formula,

$$\text{area} = \frac{1}{2}ab\sin C$$

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$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

So

$$AB = 2x - 1 = 6, \qquad AC = 2x + 1 = 8$$

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$$BC = 4.1063\ldots$$

The largest side is $AC = 8$, so the largest angle is $\angle B$.
Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\ldots^\circ$$

FINAL ANSWER

$\boxed{103^\circ}$